

Sagar Patel

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SUMMARY

Data Scientist with over 4 years of experience in data Science, deep learning, and predictive modelling. Proficient in Python, SQL, and Power BI, with a proven track record of building scalable AI solutions and delivering actionable insights. Skilled in designing end-to-end machine learning pipelines, deploying models using FastAPI and cloud platforms, and integrating APIs for intelligent automation. Experienced in working with Large Language Models (LLMs) and Generative AI technologies such as OpenAI, Gemini Pro, LLaMA, LangChain, and Huggingface Transformers. Strong hands-on expertise in Retrieval-Augmented Generation (RAG), vector databases, fine-tuning NLP models, and developing real-world generative AI applications. Detail-oriented, innovation-driven, and capable of transforming complex data into measurable business value.

WORK EXPERIENCE

Go Digit Insurance *Data Scientist Ahmadabad* Axis Bank Property

11/2024 -Present

- Objective: Automated the extraction of key-value pairs and answers from scanned PDF documents, such as Axis Bank insurance policy documents, using OCR, Azure Form Recognizer, and OpenAI-based NLP
- Customer: Digit Insurance boat team
- Tech Stack: FastAPI (Python), Cloud Services: Azure Form Recognizer (OCR), AWS S3 (File storage and retrieval), NLP/AI: OpenAI GPT-based APIs for key-value extraction,
- Role: Designed and implemented an end-to-end pipeline for automated processing of scanned documents (e.g., Axis Bank insurance policy PDFs), integrating Azure Form Recognizer for OCR, OpenAI APIs for intelligent data extraction, and FastAPI-based RESTful services for batch processing and result retrieval. The solution included robust file handling, S3 integration, error management, automated PDF splitting, image extraction, and asynchronous operations, ensuring scalability, modularity, and maintainability of the entire codebase.

Life User Declaration

- Objective: To automate the extraction, transcription, and validation of customer declarations for life insurance policies using advanced NLP and speech-to-text technologies. The system ensures accurate comparison between customer-provided information and data extracted from audio/video declarations.
- Customer: Digit Insurance
- Tech Stack: FastAPI (Python), Cloud Platform: Microsoft Azure (Key Vault, Cognitive Services Speech-to-Text), NLP: LLM-based API for information extraction, Audio Processing: pydub, fleep, Database PostgreSQL
- Role: Designed and implemented the end-to-end FastAPI backend pipeline, integrating Azure Key Vault for secure credential management, DMS integration modules, audio-to-text conversion using Azure Speech-to-Text, and NLP-based information extraction.

Topia Life sciences *Data Scientist Ahmadabad* Retrosynthesis with MCTS

01/2024 - 10/2024

- Implemented an automated retrosynthesis prediction engine within the Stemlite application, using a template-based approach for synthesis planning.
- Reaction Templates: Extracted and applied reaction templates to identify plausible disconnection strategies for organic molecules.

- Neural Network Policy Model: Developed a feed-forward neural network to rank and prioritize reaction templates based on molecular context.
- Monte Carlo Tree Search (MCTS): Employed MCTS for efficient search through possible synthesis pathways to identify optimal synthetic routes

Predictive Research sr. Data Analyst Bangalore

02/2021 - 08/2023

LEAD SCORING ENGINE

- Objective: Predict customer propensity to buy products.
- Customer: Auto data Solutions/Hyundai Motor America.
- Tech Stack: Python, Scikit-Learn, MySQL Server, Flask, Docker.
- Role: Data collection, cleaning, feature selection, model building, continuous testing and deployment, report submission, model accuracy improvement.

PREDICT-OPS

- Objective: Develop log monitoring product.
- Tech Stack: Python, Django Rest, Elasticsearch, Prometheus, EC2, MongoDB, PostgreSQL.
- Role: Data ingestion from Elasticsearch, pattern recognition, micro-service design and development, Elasticsearch-alert configuration.

CUSTOMER LIFETIME VALUE

- Objective: Identify the best customers.
- Tech Stack: Python, Scikit-Learn, SQL
- Role: Data collection, cleaning, feature selection, model building, continuous testing.

EDUCATION DETAILS

Government Engineering College Surat CGPA: 6.98

Diploma in Information Technology, [2014-2017]

Government Engineering College Gandhinagar CGPA: 7.23

Bachelor in Information Technology, [2017-2020]

Internship

Tops Technology: python

Applied AI: Artificial Intelligence

PROFESSIONAL SKILLS

- Programming Languages: Python
- Machine Learning: TensorFlow, Keras, Scikit-Learn, PyTorch
- Data Analysis: NumPy, Pandas, Matplotlib, Seaborn
- Tools: Power BI, Excel
- Web Development: Django, Flask, Streamlit
- Databases: PostgreSQL, MySQL, MongoDB
- Generative AI: Gemini Pro, OpenAI, LLaMA, Langchain, Huggingface Transformers, RAG Pipelines
- Vector Databases: Pinecone, FAISS, Chroma
- Deployment & APIs: FastAPI, RESTful APIs, Model Deployment on Cloud (Azure, AWS), CLI tools

PROJECTS

Care Predict

- Developed ML and DL models to diagnose and predict diseases like Alzheimer's, brain tumours, breast cancer, COVID-19, diabetes, heart disease, and pneumonia.
- Utilized CNNs for image recognition and Random Forests for tabular data.
- Built and deployed a Flask application for diagnostic predictions.

Buy Till You Die - Customer Lifetime Value Prediction

- Predicted customer lifetime value (CLTV) using the Beta-Gamma-NBD (BG/NBD) approach.
- Gained valuable insights into customer behaviour.

Movie Recommender System with Streamlit

- Developed a movie recommender system using cosine similarity and Streamlit.
- Analysed movie metadata to calculate similarities and provide recommendations.
- Built a user-friendly interface with Streamlit for personalized movie recommendations.

Dell Image Generation with LLM APIs

- Used LLMs and APIs to generate unique images for Dell's technology.
- Experimented with OpenAI's DALL-E 2 APIs to create Dell-specific imagery.

Objective

Seeking to leverage my expertise in Python, deep learning, machine learning, and Generative AI to design, deploy, and optimize end-to-end AI solutions. I aim to contribute to impactful projects across domains by building scalable AI pipelines, enhancing data-driven decision-making, and driving innovation through predictive modeling and automation.